Comparative Evaluation of Federated Aggregation Strategies for Classification of Chest X-Ray Images

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Abstract

Federated Learning (FL) has emerged as the best paradigm for collaborative training of deep learning models on decentralized data. This study presents a comparative evaluation of three prominent federated aggregation strategies—FedAvg, FedProx, and SCAFFOLD—for the classification of chest X-ray images into multiple disease categories, including COVID-19, Pneumonia, Tuberculosis, and Normal. We use vision transformer (ViT) as our model to train. We address the challenges from data heterogeneity and class imbalance across distributed clients using Dynamic Adaptive Focal Loss (DAFL) along with adaptive aggregation mechanism. We conduct extensive experiments on IID and non-IID datasets. We also highlight the critical role of aggregation strategies in federated learning and provides practical insights for deploying privacy-preserving AI systems in distributed medical environments.

Keywords: Federated Learning, Chest X-ray Classification, Vision Transformer, FedAvg, FedProx, SCAFFOLD, Non-IID Data, Medical Imaging, Privacy-Preserving Learning, Aggregation Strategies

1 Introduction

1.1 Background and Motivation

CNNs (Convolutional Neural Networks) are popular in computer vision for extracting spatial characteristics from images. However, their local receptive fields require deeper networks to capture broader context.

Transformer-based models, such as Vision Transformers (ViTs), have excelled in medical imaging, often matching or outperforming CNNs. Inspired by NLP Transformers, ViTs split images into patches, embed them as tokens, and use self-attention to identify long-range relationships, thereby boosting performance.[9]

Training deep learning models for medical imaging requires broad datasets, but using data from a single institution limits generalization because of differences in patient characteristics and locations. Privacy laws such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) prevent the sharing of raw patient data, impeding the combination of datasets. Federated Learning (FL) allows institutions to jointly train models by sharing model updates rather than raw data, thereby protecting privacy. This chapter explores methods for aggregating models to preserve privacy chest X-ray classification in both independent and identically distributed (IID) and non-independent and identically distributed (Non-IID) settings, attempting to develop a robust, general framework.[11]

1.2 Importance of Aggregation in Federated Learning

Model aggregation in FL combines client-trained models into one global model while preserving privacy. The chosen aggregation method greatly affects model accuracy. In centralized federated learning, model aggregation takes place at a central server. This server gathers trained model updates from each client and combines them into one improved model.

We run numerical experiments training deep neural networks (DNNs) with various datasets in IID and Non-IID settings. We study three aggregation strategies, namely: FedAvg, FedProx, SCAFFOLD.

Data Heterogeneity One of the major challenges in federated learning is data heterogeneity. In FL systems, local datasets across clients often differ significantly in terms of class distribution, data volume, sensor configurations, and feature characteristics. These differences lead to non-independent & non-identically distributed (Non-IID) data, which can negatively impact model convergence, reduce generalization, and cause instability during training.[3]

Variations in System Configurations. Clients participating in federated learning typically operate under diverse system conditions. Hence there can be variations in hardware resources, network bandwidth, storage capacity, and energy constraints which can affect the training process. As a result, adaptive client selection mechanisms are often required to select available, capable clients at each communication round.[3]

Fairness and Personalization. Optimizing a single global model for all clients may not benefit all participants equally, particularly those with minority-class distributions or unique data characteristics. This can lead to unfair performance across clients. In sensitive domains such as healthcare and finance, ensuring fairness and personalization of models becomes critically important.[3]

Scalability. Scalability remains a major challenge for the large-scale deployment of federated learning systems. Managing thousands, or even millions, of participating clients requires efficient communication protocols, scalable aggre-

gation mechanisms, and robust infrastructure to support model updates and bandwidth allocation.[3]

Privacy and Security Concerns. Despite improved privacy, FL remains vulnerable to security threats, including model inversion, gradient leakage, and malicious updates, which can compromise data and model integrity, drop out or participate unpredictably, slowing convergence and reducing stability.[3]

In non-IID settings, class-aware loss functions that adjust weights to account for local class imbalance can improve model robustness.

Aggregation of Client Updates. A central server in federated learning collects model updates from participating clients and aggregates them to improve the global model. Choosing an appropriate aggregation strategy is critical, as the method's effectiveness depends on the data distribution, particularly whether it is IID or Non-IID.[3]

1.3 Objectives and Contributions of the Chapter

This paper explains the working principles of three well-known federated learning aggregation strategies: FedAvg, FedProx, SCAFFOLD. The study examines how these strategies perform in both independent and identically distributed (IID) and non-independent & identically distributed (non-IID) data settings. It pays special attention to how each aggregation method addresses data variability, communication efficiency, and model convergence in distributed training environments.

Our main goal is to evaluate the strength and effectiveness of these federated aggregation techniques , particularly for tasks involving chest X-ray data diagnostics. By carefully examining their training behavior and performance metrics, this work seeks to identify which strategies work best in real-world healthcare situations where data is spread across multiple institutions. The findings provide insights into the advantages and disadvantages of each method and highlight their potential to enable privacy-focused collaborative learning in medical imaging applications.

- To evaluate different federated aggregation strategies for chest X-ray disease detection.
- To analyse performance of federated aggregation strategies under IID and Non-IID data distributions.
- To study communication efficiency and convergence behavior.
- To provide practical insights for deployment in healthcare systems.

2 Federated Learning Framework

2.1 Problem Formulation

We consider the task of image classification in a **Federated Learning (FL)** framework, where a global model is collaboratively trained across K distributed

clients. Each client k owns a local dataset. [36]

$$D_k = (x_{k,i}, y_{k,i})_{i=1}^{N_k}, \tag{1}$$

[36]

where $x_{k,i}$ denotes the i-th input image of k-th client and $y_{k,i} \in 1, 2, \ldots, C$ represents its corresponding class label among C possible classes. The overall training data is distributed across clients such that [36]

$$D = \bigcup_{k=1}^{K} D_k. \tag{2}$$

[36]

A key challenge in this setting is **data heterogeneity**, arising from non-identically distributed (non-IID) data across clients and class imbalance, where certain classes may have significantly fewer samples than others.

Let $f(x; \theta)$ denote the global model parameterized by θ. The objective is to learn model parameters that achieve strong classification performance across all clients and classes while respecting the decentralized nature of the data.

The global optimization problem is formulated as minimizing a **weighted aggregation of local empirical losses**:[36]

$$\min_{\theta} \sum_{k=1}^{K} \alpha_k^{(\cdot)}, \mathcal{L}_k(\theta; D_k), \tag{3}$$

[36] where

- $\mathcal{L}_k(\theta; D_k)$ denotes the local loss computed on client k's dataset,
- $\alpha_k^{(\cdot)}$ represents the **dynamic weighted averaging coefficient** assigned to client k, [6]
- the formulation intentionally leaves the weighting function **open for dynamic adaptation** (e.g., based on data quality, class distribution, training progress, or other criteria).

The aggregation weights satisfy the normalization constraint.

$$\sum_{k=1}^{K} \alpha_k^{(\cdot)} = 1, \qquad \alpha_k^{(\cdot)} \geq 0. \tag{4}$$

This flexible formulation enables the integration of **dynamic weighting strategies** that can adapt during training to mitigate performance degradation caused by heterogeneous client data distributions.

To capture global contextual relationships in images, we adopt a **Vision Transformer (ViT)** architecture. ViT represents an image as a sequence of flattened patches and employs a transformer encoder to model long-range dependencies between image regions. This design allows the model to dynamically attend to informative parts of the image, thereby improving representation learning under heterogeneous data conditions.

2.2 Objective Function

In federated learning, the global model is trained collaboratively across multiple decentralized clients without sharing their raw data. However, real-world federated datasets often exhibit significant class imbalance and heterogeneity across clients. To address this challenge, the proposed framework introduces a dual-level imbalance modelling mechanism consisting of a *client-level imbalance factor* and a *federation-level imbalance factor*. These imbalance signals are incorporated into a Dynamic Adaptive Focal Loss (DAFL) [36] used during client-side training. Furthermore, an adaptive weighted aggregation strategy is employed at the server to combine the locally trained models.

The overall global objective of the federated system is defined as the minimization of the expected loss across all participating clients:[36]

$$\min_{w} F(w) = \sum_{k=1}^{K} \frac{N_k}{N} \mathbb{E}(x, y) \sim D_k \left[-(1 + cf, y)(1 - p_y)^{\gamma} \log(p_y) \right] \tag{5}$$

where K denotes the number of clients, N_k represents the number of samples at client k, and $N = \sum_{k=1}^{K} N_k$ is the total number of samples across the federation. D_k denotes the local dataset of client k, and p_y is the predicted probability of the true class y. The term $c_{f,y}$ represents the federation-level imbalance coefficient associated with class y.

Client-Level Imbalance

The imbalance within each client dataset is quantified using a client-level imbalance parameter (c_k), which measures the skewness of the class distribution at client (k). It is defined as [36]

$$c_k = \frac{1}{C} \sum_{i=1}^{C} \frac{N_k - n_{k,i}}{n_{k,i} + \epsilon} \tag{6}$$

where C denotes the total number of classes, $n_{k,i}$ represents the number of samples of class i at client k, and ϵ is a small constant added for numerical stability.

Federation-Level Imbalance

To capture the imbalance across the entire federation, a global imbalance coefficient $c_{f,i}$ is computed at the server for each class i. This coefficient highlights classes that are globally underrepresented across all participating clients, enabling the model to focus more strongly on rare classes. [36]

$$c_{f,i} = \frac{N - N_i}{N_i + \epsilon} \tag{7}$$

where (N_i) represents the total number of samples belonging to class (i) across all clients.

Dynamic Adaptive Focal Loss

During client-side training, the model parameters are optimized using a Dynamic Adaptive Focal Loss (DAFL). This loss function integrates the federation-level imbalance coefficient into the focal loss formulation, thereby simultaneously emphasizing hard examples and rare classes.

For a sample with true class (t), the loss is defined as [36]

$$\mathcal{L}DAFL = -(1 + cf, t)(1 - p_t)^{\gamma} \log(p_t) \tag{8}$$

where p_t denotes the predicted probability of the true class t, and γ is a focusing parameter that controls the contribution of hard versus easy examples.

Training Objective

Each client optimizes its local objective using the DAFL loss over its dataset. The local objective function for client (k) is defined as

$$F_k(w) = \frac{1}{|D_k|} \sum_{(x,y)\in D_k} \mathcal{L}_{DAFL}(x, y) \tag{9}$$

where $|D_k|$ denotes the number of samples in the local dataset of client k.

Side Model Aggregation

After local training, the updated client models are transmitted to the server, where they are aggregated to update the global model. The aggregation is performed using adaptive weighted averaging:

$$w^{t+1} = \sum_{k=1}^{K} \alpha_k w_k^t \tag{10}$$

where w_k^t represents the model parameters obtained from client k at communication round t, and α_k denotes the aggregation weight assigned to client k. These weights can be designed to reflect the size or imbalance characteristics of the client data.

This formulation enables the federated model to simultaneously account for local data heterogeneity, rare global classes, and varying sample importance across the federation, thereby improving robustness in highly imbalanced federated learning environments.

3 Role of Aggregation in Model Convergence

Aggregation is a crucial part of the federated learning (FL) framework. In a distributed learning setting, each client trains a model using its local dataset. The aggregation process combines the knowledge gained by all participating clients into one global model. This is similar to collaborative learning, where information from various decentralized datasets is combined to form a global model.

The main goal of aggregation is to ensure that the global model reflects patterns across different data sources without requiring the transfer of raw data. Since federated learning often involves diverse data distributions and varying

client resources, several aggregation algorithms have been proposed to effectively handle different training scenarios. In ideal cases where client datasets follow an Independent and Identically Distributed (IID) distribution, simple aggregation strategies often work well. One popular algorithm used in these situations is Federated Averaging (FedAvg). FedAvg aggregates the locally trained client models by averaging their parameters. The weights correspond to the number of training samples each client has compared to the total number of samples across all clients. However, real-world federated learning environments seldom meet the IID assumption. Often, datasets from different clients exhibit Non-IID distributions, meaning that data characteristics, class distributions, or feature representations can vary significantly between organizations. This heterogeneity can lead to client drift, slower convergence, and reduced model performance when using simple aggregation methods. To tackle these issues, several improved aggregation algorithms have been introduced.

Several aggregation algorithms are commonly used in federated learning, including:

- **FedAvg:** FedAvg Performs weighted averaging of client model parameters or gradients, where the weights are determined by the proportion of local training samples at each client.
- **FedProx:** FedProx is an extension of FedAvg where we incorporate a proximal regularization term in the local objective function to limit excessive deviation of local updates from the global model.
- **SCAFFOLD:** Introduces control variates to mitigate client drift and improve convergence in federated settings with non-IID data.

These aggregation algorithms help federated learning systems effectively combine knowledge from multiple clients while ensuring data privacy and improving model performance across various data sources.

4 Aggregation Strategies Considered

4.1 FedAvg

Federated Averaging (FedAvg) is a widely used optimization algorithm in federated learning. In this setting, the local objective function corresponding to the global objective at device k is denoted by F_k. Each participating device employs stochastic gradient descent (SGD) as its local optimization method, using the same learning rate and number of local epochs. [25]

Algorithm: Federated Averaging

Input: $K, T, \eta, E, w^0, N, p_k, ; k = 1, \ldots, N$

for $t = 0, \ldots, T-1$ **do**

Server selects a subset S_t of K devices at random (each device k is chosen with probability p_k)

Server sends w^t to all selected devices

for each device $k \in S_t$ **in parallel do**
Perform E epochs of SGD on local objective F_k with learning rate η
Obtain updated model parameters w_k^{t+1}
Send w_k^{t+1} to the server
end for
Server aggregates the received models

$$w^{t+1} = \frac{1}{|S_t|} \sum_{k \in S_t} w_k^{t+1}$$

end for

During each communication round, a subset K of the total devices is selected to perform local training. Each selected device runs SGD on its local dataset for E epochs. After completing the local updates, the resulting model parameters from all participating devices are sent to the server and aggregated through weighted averaging.

This aggregation strategy performs effectively when multiple local epochs are used. Increasing the number of local epochs enables more local computation and reduces communication frequency, thereby accelerating convergence in networks with limited communication. However, when the local objective functions F_k differ substantially across devices, performing many local epochs may cause each device to drift toward a solution optimal for its local data rather than the global objective. The FedAvg algorithm is as follows [25]

4.2 FedProx

The FedProx algorithm [25] extends the Federated Averaging (FedAvg) by introducing a proximal regularization term into the local optimization objective. This modification is designed to address the challenges caused by statistical heterogeneity among distributed clients, which is common in federated environments where data distributions differ significantly across institutions. In Non-IID scenarios, performing multiple local updates can lead to substantial divergence between local client models and the global model. The proximal term constrains the local optimization process by penalizing large deviations from the current global model parameters. As a result, client updates remain closer to the shared global model, improving training stability and convergence behavior. In this study, the local objective function integrates the proximal regularization term with the Dynamic Adaptive Focal Loss employed chest X-ray classification. This combination enhances robustness when training across clients with imbalanced class distributions and varying dataset sizes, which are typical characteristics of medical imaging datasets. Additionally, a dynamic aggregation strategy is implemented at the server to improve the reliability of model updates. Unlike conventional FedAvg-style aggregation that relies solely on dataset-size weighting, the proposed method computes adaptive aggregation weights based on three client-specific indicators: the size of the local dataset, the local validation accuracy, and the magnitude of model drift from the global parameters.

FedProx algorithm is as follows [25]

Algorithm: FedProx with Dynamic Weighted Aggregation

Input: $K, T, \mu, \gamma, w^0, N, p_k, ; k = 1, \ldots, N$

for $t = 0, \ldots, T-1$ **do**

Server selects a subset S_t of K devices at random
(each device k is chosen with probability p_k)

Server sends w^t to all chosen devices

for each device $k \in S_t$ **in parallel do**

Device k computes local model update

$$w_k^{t+1} \approx \arg\min_w \; F_k(w) + \frac{\mu}{2}\|w - w^t\|^2$$

Compute local metrics

Data size n_k

Local accuracy a_k^t

Model drift $d_k^t = \|w_k^{t+1} - w^t\|$

Send $\{w_k^{t+1}, n_k, a_k^t, d_k^t\}$ to server

end for

Server computes dynamic aggregation weights

$$\alpha_k^t = \frac{f(n_k, a_k^t, d_k^t)}{\sum_{j \in S_t} f(n_j, a_j^t, d_j^t)}$$

Server aggregates client models

$$w^{t+1} = \sum_{k \in S_t} \alpha_k^t w_k^{t+1}$$

end for

4.3 SCAFFOLD

The SCAFFOLD (Stochastic Controlled Averaging for Federated Learning) [20] algorithm is designed to address the problem of *client drift* that arises in federated learning systems with heterogeneous data distributions. In traditional federated optimization methods such as FedAvg, local updates on non-identically distributed (Non-IID) client data may diverge from the global optimization direction, slowing convergence and reducing model stability. SCAFFOLD mitigates this issue by introducing *control variates* that estimate and correct the drift between client updates and the global optimization trajectory. Specifically, the server maintains a global control variate c, while each client maintains its own local control variate c_i. These variables act as estimates of the expected update direction of the global model and the local client models, respectively. During local training, each client adjusts its gradient update using the correction term $(c - c_i)$. This correction compensates for the deviation introduced by heterogeneous local datasets and ensures that local updates remain aligned with the global optimization direction. After completing the local optimization steps, clients compute

both model updates and control variate updates, which are transmitted to the server for aggregation. At the server side, the received updates are averaged across participating clients to update both the global model parameters and the global control variate. This coordinated update mechanism effectively reduces the variance caused by client heterogeneity and improves the stability of the federated training process. From an optimization perspective, SCAFFOLD can be viewed as a *variance-reduction technique* for federated learning. By reducing the variance introduced by heterogeneous client updates, the algorithm significantly improves convergence speed and requires fewer communication rounds compared to traditional aggregation methods.

Algorithm: SCAFFOLD aggregation algorithm

Server input: initial model x and control variate c, global step-size η_g
Client input: c_i and local step-size η_l
for each round $r = 1, \ldots, R$ **do**
  Sample clients $S \subseteq 1, \ldots, N$
  Communicate (x, c) to all clients $i \in S$
  for each client $i \in S$ **in parallel do**
    Initialize local model $y_i \leftarrow x$
    for $k = 1, \ldots, K$ **do**
      Compute stochastic gradient $g_i(y_i)$
      $y_i \leftarrow y_i - \eta_l(g_i(y_i) - c_i + c)$
    end for
    Update control variate

$$c_i^+ \leftarrow c_i - c + \frac{1}{K\eta_l}(x - y_i)$$

    Compute model and control updates

$$(\Delta y_i, \Delta c_i) \leftarrow (y_i - x, c_i^+ - c_i)$$

    Send $(\Delta y_i, \Delta c_i)$ to server
    $c_i \leftarrow c_i^+$
  end for
  Aggregate updates

$$(\Delta x, \Delta c) \leftarrow \frac{1}{|S|}\sum_{i \in S}(\Delta y_i, \Delta c_i)$$

  Update global variables

$$x \leftarrow x + \eta_g \Delta x$$

$$c \leftarrow c + \frac{|S|}{N}\Delta c$$

end for

5 Experimental Methodology

5.1 Chest X-Ray Dataset Description

The performance of deep learning models, particularly in medical image analysis, largely depends on the quality and diversity of the training data. In this study, publicly available chest X-ray datasets containing multiple respiratory disease categories are utilized to simulate a distributed healthcare environment. These datasets allow us to evaluate the capability of federated learning algorithms to collaboratively learn diagnostic patterns across multiple institutions while preserving data privacy. The experiments were conducted using two publicly available chest X-ray image datasets containing radiographic images of various respiratory diseases. The datasets consisted of chest X-ray images of four classes: COVID-19, Pneumonia, Tuberculosis, and Normal. These datasets were selected to simulate real-world healthcare scenarios in which hospitals maintain their own patient records and cannot share raw medical data due to privacy regulations. We then divided this dataset into a train, a validation, and a test set for training and validation.

5.2 Data Preprocessing and Augmentation

Preprocessing plays a particularly important role in federated learning environments, where data is distributed across multiple clients or institutions. In such settings, each client may collect medical images using different imaging devices, acquisition protocols, or clinical conditions, resulting in variations in image quality and feature representation. If these inconsistencies are not addressed through preprocessing, the local models trained at different clients may learn significantly different feature distributions. This can lead to unstable model updates during federated aggregation and degrade the global model's performance. We use preprocessing techniques across all clients to ensure that the input data is standardized, reducing discrepancies between local datasets and enabling the model to learn more generalizable patterns. In addition to standardization, data augmentation techniques are also employed to artificially increase the diversity of the training data. Augmentation methods, such as random rotations and horizontal flips, help the model learn invariant features that are robust to variations in image orientation and position. The preprocessing pipeline implemented in PyTorch is shown below:

```
_transform = transforms.Compose([
transforms.Resize((224, 224)),
transforms.RandomHorizontalFlip(),
transforms.ToTensor(),
transforms.Normalize([0.485, 0.456, 0.406],
[0.229, 0.224, 0.225])
])
```

5.2.1 Image resizing

As a first step, all images were resized to a fixed spatial resolution of 224 $\times$ 224 pixels. Standardizing the image size is an important preprocessing step because deep learning architectures require inputs of consistent dimensions. On resizing the images to a uniform resolution, the model can process the input data efficiently without encountering dimensional mismatches during training. Furthermore, selecting a resolution of 224 $\times$ 224 pixels aligns with the input requirements of pretrained vision transformer model used and allows the model to capture sufficient spatial information from the chest X-ray images while maintaining computational efficiency.

5.2.2 Image Normalization

Following image resizing, normalization was applied to standardize the pixel intensity values across the dataset. The images were normalized using the widely adopted ImageNet mean and standard deviation values, given by: Mean = [0.485, 0.456, 0.406] Standard Deviation = [0.229, 0.224, 0.225] Normalization transforms pixel values to a standardized range, helping stabilize the training process and improving the numerical behavior of optimization algorithms. In deep learning, especially when training large neural networks, normalization reduces internal covariate shift and enables faster convergence during gradient-based optimization. As a result, the model can learn more effectively from the input data and achieve improved generalization performance. In the context of federated learning, consistent preprocessing across all participating clients is particularly important. Since the data is stored locally at each client, applying identical preprocessing procedures ensures that the local models are trained on inputs with the same statistical distribution. This consistency helps reduce discrepancies in client updates during model aggregation and contributes to a more stable, reliable global model.

5.3 Model Architecture for Chest X-ray Classification

The model used for this study is a Vision Transformer whose architecture [36] is as follows:

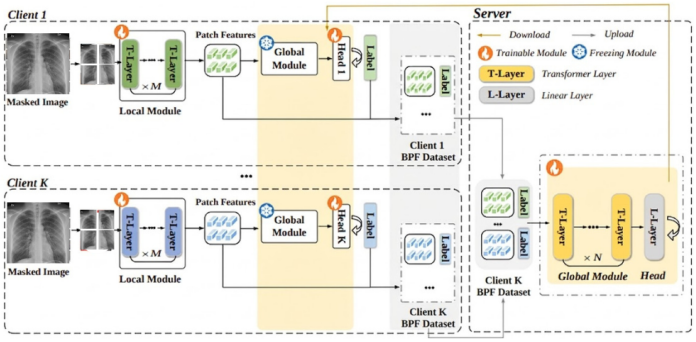


Figure 1: ViT pretrained model in the client and the server [36]

The proposed architecture utilizes the ViT-Small Patch16-224 model, which divides the input image into smaller patches and processes them using a transformer encoder. This design enables the model to learn both local and global visual representations, which is particularly beneficial for identifying subtle patterns present in chest X-ray images. The model is initialized with pre-trained weights, enabling it to leverage knowledge from large-scale image datasets. Transfer learning significantly improves performance on medical datasets, which are often smaller than general-purpose image datasets. Within the architecture, the input image of size 224×224 pixels is divided into non-overlapping 16×16-pixel patches. Each patch is then embedded into a high-dimensional feature vector, which serves as the input to the transformer encoder layers. These encoder layers use multi-head self-attention to capture relationships across different regions of the image. This allows the model to focus on important anatomical structures that may indicate the presence of respiratory diseases. To improve generalization and reduce overfitting, regularization techniques are incorporated into the model. Specifically, a dropout rate of 0.1 and a stochastic depth (drop path rate) of 0.1 are applied during training. These mechanisms help the model learn more robust feature representations while preventing over-reliance on specific neurons. In addition, the original classification head of the Vision Transformer is modified to adapt the model to the specific task of multi-class disease classification. The final classification layer is replaced with a custom head comprising a dropout layer followed by a fully connected linear layer that maps the extracted features to the four target classes: COVID-19, Pneumonia, Tuberculosis, and Normal. Further-

more, a learnable parameter γ (gamma) is introduced within the architecture to support the custom loss formulation used during training. This parameter controls the loss function's focus, enabling the model to place greater emphasis on difficult or misclassified samples during optimization. During the forward pass, the input image is first processed by the transformer feature-extraction layers. The resulting feature embeddings are normalized, and the classification token (CLS) is extracted to represent the image as a whole. This token summarizes the contextual information captured across all image patches and is then passed through the modified classification head to generate the final prediction. By combining the strengths of transformer-based architectures with customized classification layers, the proposed model effectively captures complex patterns in chest X-ray images, enabling accurate detection and classification of respiratory diseases within a federated learning framework.

5.4 Federated Setup

While the Vision Transformer architecture described in the previous section can effectively extract meaningful features from chest X-ray images, the primary objective of this research is to evaluate its performance within a federated learning framework. Unlike traditional centralized machine learning approaches, in which all data is collected and processed in a single location, federated learning enables multiple participants to collaboratively train a shared global model while keeping their data locally. The following subsection describes the configuration of the federated learning setup used in this study. **Clients**
In the proposed experimental setup, the federated system consists of two clients, each corresponding to a different chest X-ray dataset. These datasets simulate a scenario in which multiple healthcare institutions collaborate to train a shared diagnostic model. Each client maintains its own local copy of the global model and performs training using its locally available data. Since the datasets originate from different sources, the distribution of samples across clients may vary, which reflects real-world data heterogeneity commonly observed in healthcare systems. **Communication Rounds**
Federated learning training is performed over several communication rounds. During each communication round, the following steps are executed: The central server distributes the global model to all participating clients. Each client trains the received model locally using its private dataset. After completing local training, the clients send their updated model parameters back to the server. The server aggregates the updates received from all clients to produce an improved global model. This process is repeated for a fixed number of communication rounds to gradually improve the global model's performance. In this study, the federated training process was conducted over 20 communication rounds, allowing the model to progressively learn from distributed datasets while maintaining training stability.We also used EarlyStopping to prevent overfitting

Local Epochs
During each communication round, clients perform multiple local training epochs on their respective datasets before transmitting the model updates to the server.

Local epochs determine how many times the client iterates over its dataset during a single communication round. Using multiple local epochs allows each client to extract richer feature representations from its local data before contributing updates to the global model. However, increasing the number of local epochs may also introduce client drift, particularly when data distributions differ significantly across clients. Each client performs 2 local epochs per communication round in the experimental setup, to balance learning efficiency and training stability . This configuration allows the model to learn sufficiently from local datasets while maintaining effective global aggregation.

Global Model Aggregation

After receiving the updated model parameters from all clients, the server aggregates the updates using different federated optimization algorithms, including FedAvg, FedProx,SCAFFOLD. These algorithms differ in the way they handle model updates and mitigate issues such as client drift and data heterogeneity. The aggregated global model is then redistributed to the clients for the next communication round, enabling continuous collaborative learning across the distributed datasets. Through this iterative process of local training and global aggregation, the federated learning framework enables the development of a robust disease classification model without requiring centralized access to sensitive medical data.

Table 1: Training Hyperparameters

Parameter & **Value**	
Global Communication Rounds	25
Local training epochs	2
Batch size	64
Learning rate	$1e^{-4}$
Optimizer	AdamW
Weight decay	$5e^{-3}$
Gradient clipping	1.0
LR scheduler	ReduceLROnPlateau
EarlyStopping	Patience = 3 or 7

5.5 Data Partitioning Strategy

5.5.1 IID (Independent and Identically Distributed) Partitioning

In the IID setting, the dataset is distributed across clients such that each client receives a similar proportion of samples from every class. This means the dataset's statistical properties remain consistent across all participating clients. Under this configuration, each client observes a balanced dataset, allowing locally trained models to learn similar feature distributions. This scenario represents an idealized federated learning environment where the local updates generated by different clients are relatively consistent with one another. As a result, when these updates are aggregated at the central server, the global model converges smoothly and efficiently. Evaluating federated learning algorithms under IID

conditions provides a baseline for understanding their performance in environments with minimal data heterogeneity.

5.5.2 Non-IID Partitioning Using Dirichlet Distribution

We use the Dirichlet distribution [14] to simulate realistic federated learning scenarios where clients have heterogeneous data distributions. For each class in the dataset, all files are collected and randomly shuffled. A Dirichlet distribution with parameter α generates a probability vector that determines how samples of that class are distributed among the clients. The parameter α controls the degree of heterogeneity in the data distribution:

- Small α values (e.g., $\alpha = 0.5$) produce highly skewed distributions, resulting in strongly Non-IID data.
- Larger α values produce more balanced distributions, making the data closer to IID.

The sampled Dirichlet proportions are multiplied by the total number of files in the class to determine how many samples each client receives. To address rounding discrepancies, adjustments are applied to ensure the total number of assigned files matches the original dataset size. The selected files are then copied to each client's class directory. This method produces heterogeneous client datasets, in which some clients may contain significantly more samples from specific classes than others. Such a setup effectively reflects the data imbalance and distribution shifts commonly observed in real-world federated learning environments.

5.6 Evaluation Metrics

5.6.1 Evaluation Metrics and Performance Analysis

We use a systematic approach for evaluating the performance of the trained models. Since federated learning involves distributed training across multiple clients and iterative aggregation at the central server, it is important to evaluate not only the predictive performance of the global model but also the efficiency of the training process.

To analyze the performance of the proposed federated learning framework, several evaluation metrics are used. These metrics provide a comprehensive understanding of the model's classification performance and the efficiency of the federated training process. Since the task involves multi-class classification of chest X-ray images into four disease categories, it is important to evaluate the model using metrics that capture different aspects of prediction quality.

In the federated learning setting, evaluation is performed using the global model obtained after each communication round. Once the server aggregates the local model updates from all clients, the updated global model is evaluated on a combined test dataset containing samples from all participating clients. This evaluation strategy ensures that the model's performance reflects its ability to generalize across distributed, heterogeneous datasets.

The following metrics are used to assess the performance of the model:
Accuracy

Accuracy measures the overall proportion of correctly classified samples among the total number of predictions made by the model. It provides a general indication of the model's ability to correctly identify disease categories in chest X-ray images.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Training (Loss & Accuracy)
The performance of client models during Federated Training rounds is measured using training loss and accuracy. Steadily increasing training accuracy denotes the classification performance of the local model on the training data, reflecting stabilized aggregation. A steadily decreasing training loss indicates that the model has been successfully optimizing the objective function (Dynamic Adaptive Focal Loss in our case). **Validation (Loss & Accuracy)**
Validation metrics help in evaluating how well the global model generalizes across clients. Validation metrics are computed from the global model's performance on images not used during training. A higher validation accuracy denotes a good aggregation method. Validation metrics help with hyperparameter tuning, such as the learning rate, client participation rate, number of local epochs, and aggregation algorithm parameters.
Test (Loss & Accuracy)
Test metrics are used to observe the performance of the final aggregated model on completely unseen data. Test metrics reflect the performance of the global model in the real world.

A higher accuracy value indicates better overall classification performance.
Precision

Precision measures the proportion of correctly predicted positive samples among all samples the model predicts as positive. In medical diagnosis, precision is particularly important because it indicates how reliable the model's positive predictions are.

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

A high-precision value indicates that the model produces fewer false-positive predictions.

Recall

Recall, also known as sensitivity, measures the model's ability to correctly identify all relevant positive samples. In medical image analysis, recall is crucial because failing to detect a disease case can have serious clinical implications.

$$Recall = \frac{TP}{TP + FN} \tag{13}$$

Higher recall indicates that the model successfully identifies a larger proportion of actual disease cases.

F1-Score

The F1-score provides a balanced evaluation of the model by combining precision and recall into a single metric. It is particularly useful when dealing with imbalanced datasets, which are common in medical imaging applications.

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

This metric provides a more comprehensive assessment of the model's classification performance, accounting for both false positives and false negatives.

Communication Cost

In addition to predictive performance, federated learning systems must also consider the efficiency of communication between clients and the central server. Since model parameters are exchanged in each communication round, communication overhead can significantly affect the scalability of the federated learning framework.

Communication cost [33] is evaluated based on the number of communication rounds required for the global model to converge and achieve stable performance. Algorithms that reach high accuracy in fewer communication rounds are considered more communication-efficient.

Confusion matrix

A table used to evaluate the performance of a classification model by comparing the actual values with the predicted values. It shows us the quantity of data that was predicted correctly and incorrectly.

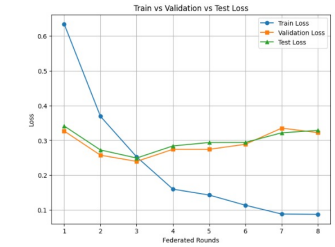
6 Results and Comparative Analysis

6.1 Performance under IID Distribution

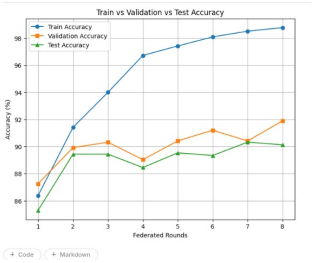
6.1.1 Performance of FedAvg under IID Distribution

FedAvg demonstrates a steady increase in training accuracy across federated rounds. The model shows good generalization under IID conditions, with a lower gap between train and validation accuracy. Divergence observed in later rounds signals mild overfitting. Nevertheless, the model shows robust test performance. The persistent gap between advanced federated aggregation strategies such as FedProx and SCAFFOLD can improve generalization and stabilize convergence.

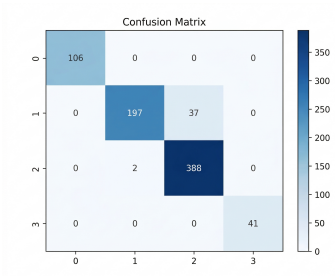
Batch size 32



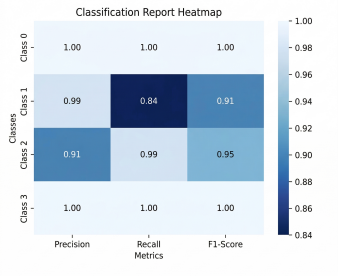
(a) Loss



(b) Accuracy

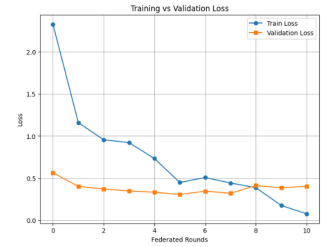


(c) Confusion matrix

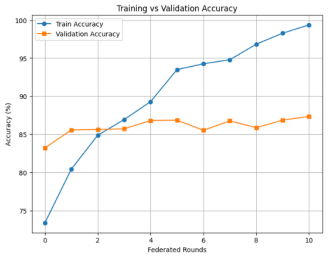


(d) Classification report

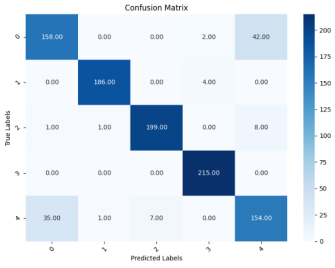
Batch size 64



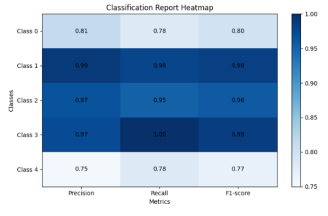
(e) Loss



(f) Accuracy



(g) Confusion matrix



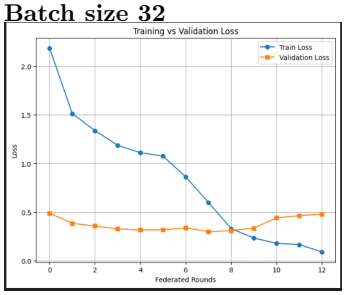
(h) Classification report

Figure 2: FedAvg performance under IID data distribution for batch sizes 32 and 64

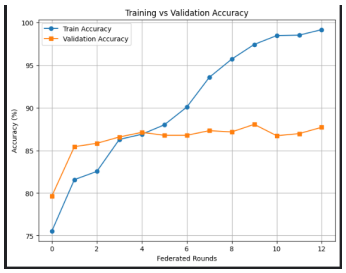
The confusion matrix also shows that most samples from classes class-0 and class-3 were correctly classified, while other classes were slightly misclassified.

6.1.2 Performance of FedProx under IID Distribution

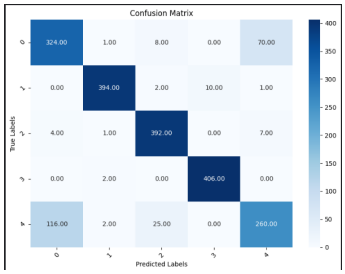
Batch size 32



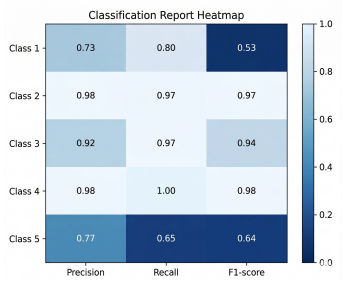
(a) Loss



(b) Accuracy



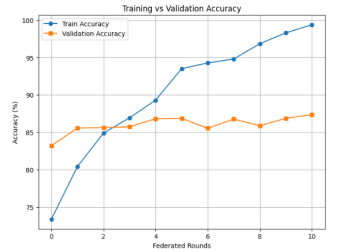
(c) Confusion matrix



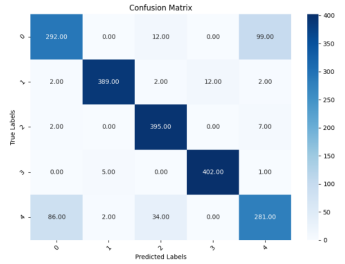
(d) Classification report

Batch size 64

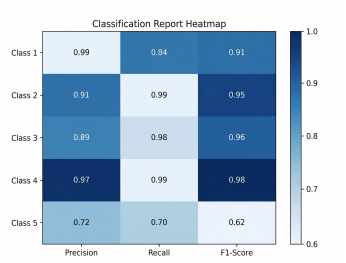
(e) Loss



(f) Accuracy



(g) Confusion matrix



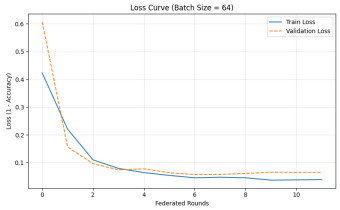
(h) Classification report

Figure 3: Fedprox performance under IID data distribution for batch sizes 32 and 64

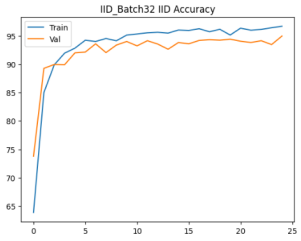
We observe that the validation accuracy reaches a slightly higher level with a batch size of 64 and exhibits a smoother curve, indicating greater convergence stability. The training accuracy curve follows a similar upward trend, reaching near-perfect values in both cases. The gap between training and validation accuracy is slightly reduced with batch size 64 compared to 32, indicating improved generalization, whereas with batch size 32.
With a batch size of 64, the training loss decreases more smoothly, and the validation loss remains more stable, indicating better optimization stability and generalization. In contrast, batch size 32 shows more fluctuation in validation loss and a clearer divergence from training loss.
As shown in the confusion matrices, most predictions lie along the diagonal, indicating correct classifications, with minor confusion observed among a few class pairs. These misclassifications are slightly reduced with a batch size of 64.
From the classification reports, we can see that most classes achieve high precision and recall, particularly the dominant classes. However, one class consistently shows lower precision and recall, indicating it is more difficult to classify. This issue is slightly mitigated with batch size 64, where class-wise performance becomes more balanced.

6.1.3 Performance of SCAFFOLD under IID Distribution

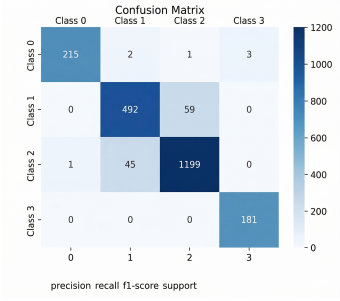
Batch size 32



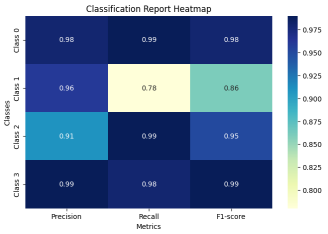
(a) Loss



(b) Accuracy

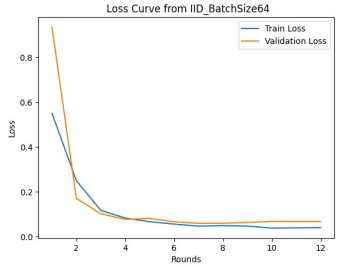


(c) Confusion matrix

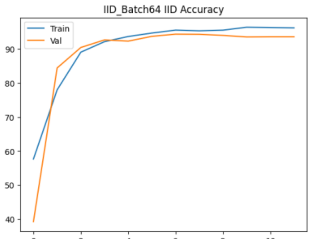


(d) Classification report

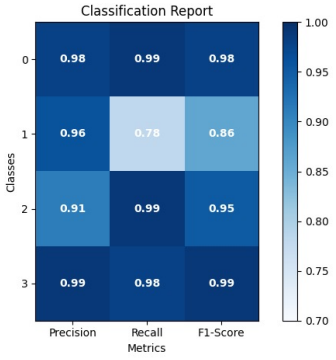
Batch size 64



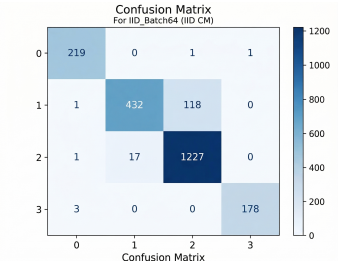
(e) Loss



(f) Accuracy



(g) Classification report



(h) Confusion matrix

Figure 4: SCAFFOLD performance under IID data distribution for batch sizes 32 and 64

The figures illustrates performance of Scaffold algorithm on IID data. We can see that Scaffold converges very quickly for IID data. We observe that the loss curves converge rapidly with generalization, with a minimal gap between the loss and accuracy curves in batch 64. With a batch size of 32, we achieve higher accuracy and lower loss, but with higher variance due to frequent updates.

The confusion matrices show significant class-wise performance compared to the non-IID setting. While there is still minor confusion between classes 1 and 2, it is considerably less.

The classification report shows balanced precision, recall, and F1 Scores across classes. Class-0 achieves strong precision and recall, while other classes exhibit higher precision and slight degradation in recall.

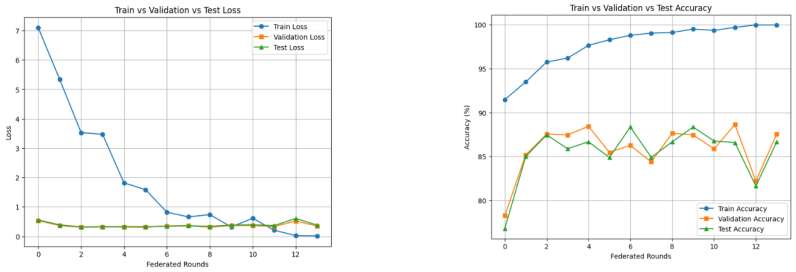
Overall, we can say that SCAFFOLD performs exceptionally well on IID data, achieving faster convergence, lower loss, and consistent performance.

6.1.4 Comparative Analysis of Algorithms under IID Distribution

The experimental results under IID data distribution demonstrate that all three federated learning algorithms—FedAvg, FedProx, and SCAFFOLD—achieve strong classification performance for the chest X-ray disease detection task. Among the evaluated methods, SCAFFOLD shows the fastest convergence and the highest validation accuracy, which can be attributed to its use of control variates that correct client updates during local training. This mechanism helps reduce variance during aggregation and accelerates global model convergence.

6.2 Performance under Non-IID Distribution

6.2.1 Performance of FedAvg under Non-IID Distribution



(a) Loss (b) Accuracy

Figure 5: FedAvg performance under non-IID data distribution

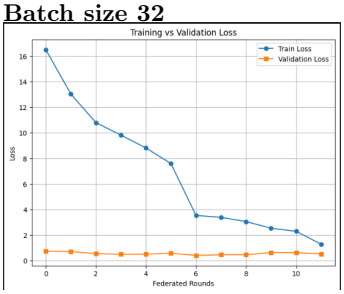
The validation accuracy and training accuracy curves for non-IID dataset depicts a large gap between validation accuracy showing unstable convergence, lack of regularization and weak generalization across clients. The model directly minimizes the loss on the training dataset using gradient-based optimization but the validation loss remains stable and not much improvement is seen in later training rounds as well. The divergence between training and validation/test performance

can largely be attributed to the heterogeneity introduced by the Non-IID data distribution. Since each client possesses datasets with varying class distributions, the locally trained models may learn different feature representations. When heterogeneous updates are aggregated using the FedAvg algorithm, the global model may experience inconsistent optimization directions, leading to fluctuations in validation and test performance.

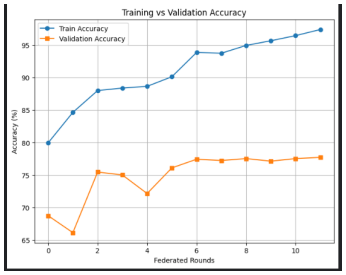
Overall, these results highlight a well-known limitation of the FedAvg algorithm in heterogeneous federated learning environments. Although the algorithm achieves high training accuracy, client drift and inconsistent local updates can degrade the generalization of the global model when client data distributions are highly diverse.

6.2.2 Performance of FedProx under Non-IID Distribution

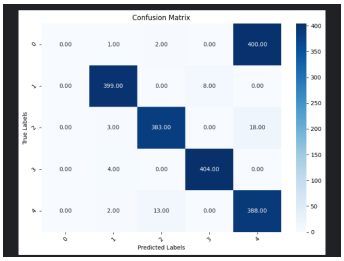
Batch size 32



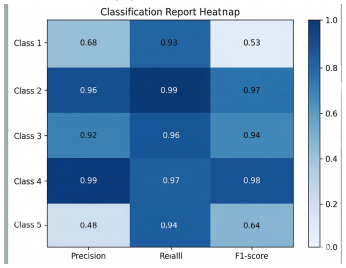
(a) Loss



(b) Accuracy

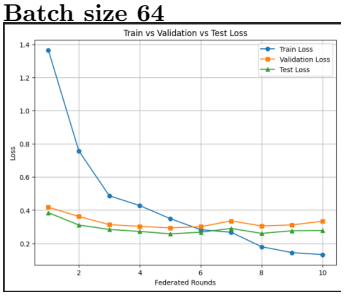


(c) Confusion matrix

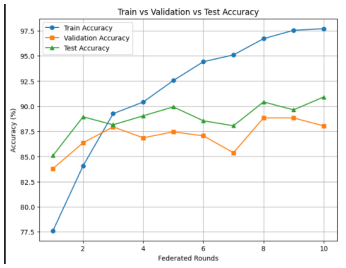


(d) Classification report

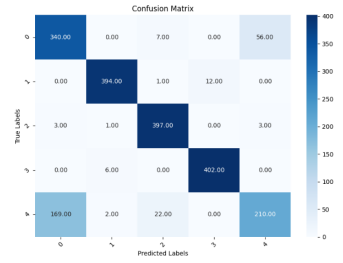
Batch size 64



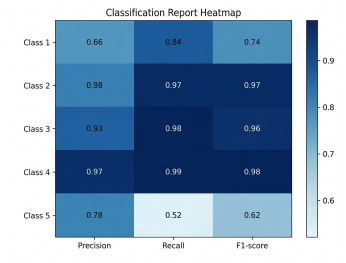
(e) Loss



(f) Accuracy



(g) Confusion matrix



(h) Classification report

Figure 6: FedProx performance under non-IID data distribution for batch sizes 32 and 64

We can observe that the validation accuracy reaches a higher level of 92% with batch size 64, exhibiting a smoother and more stable curve compared to batch size 32. The training accuracy and validation accuracy is observed to have lower gap between them for batch size 64 rather than batch size 32, suggesting instability and mild underfitting for batch size 32.
We also observe that validation and training loss decrease steadily for batch size

64, showing steady convergence. For batch size 32 we see that the validation loss remains relatively flat indicating weaker learning dynamics. From the confusion matrices we see that most classes are precisely classified with few errors in batch size 64 unlike in batch size 32.
From the classification reports we see that the precision, recall and F1 score across classes are balanced while classes 1 and 4 show differences for batch size 64 . While certain classes (especially the last one) still show relatively lower recall, overall performance is more consistent. In batch size 32, precision and recall vary more significantly across classes, with some classes experiencing reduced F1-scores due to misclassifications.

6.2.3 Performance of SCAFFOLD under Non-IID Distribution

Figure 5.6 shows SCAFFOLD's performance on non-IID data. SCAFFOLD specifically addresses client drift in federated learning caused by heterogeneous data by introducing control variates, which guide local model updates towards the global optimization direction.

Figure 7: SCAFFOLD performance under non-IID data distribution for batch sizes 32 and 64

We observe that the validation accuracy reaches a higher level with a batch size of 64 and exhibits a smoother curve, indicating greater convergence stability. The training accuracy curve follows a similar trend. The training and validation accuracies are very close with a batch size of 64, compared to 32, indicating very good generalization, whereas with a batch size of 32, we see the model underfitting.

The loss curves further confirm the effectiveness of the SCAFFOLD algorithm. In batch 64, we observe better generalization than with batch size 32, where we observe underfitting.

As shown in the confusion matrix, there is confusion between classes 1 and 2, with the other classes having the fewest confusions.

From the classification report, we can see that the model accurately classified class 0, while other classes had low precision. We can see that classes 1 were mildly misclassified, while classes 2 and 3 were severely misclassified into other classes.

Overall, the results demonstrate that SCAFFOLD achieves faster convergence, lower validation loss, and more stable accuracy, albeit at the cost of higher recall.

6.3 Convergence Analysis (Accuracy vs Communication Rounds)

In federated learning, convergence refers to the global model's ability to progressively improve its predictive performance as the number of communication rounds between clients and the server increases.The convergence patterns observed in the experimental results indicate that the three algorithms exhibit different learning behaviors across communication rounds. In general, the training accuracy for all algorithms increase steadily,however, the rate of improve-

ment in validation accuracy varies across the algorithms. The FedAvg algorithm converges more slowly and less stably under heterogeneous data distributions In contrast, FedProx demonstrates more stable convergence behavior. By incorporating a proximal regularization term in the local optimization objective, FedProx restricts the deviation of local model updates from the global model parameters. Among the evaluated methods, SCAFFOLD exhibits the most consistent convergence pattern. The algorithm effectively corrects the direction of local updates using control variates, thereby mitigating client drift caused by heterogeneous data distributions. As a result, the accuracy curve stabilizes more quickly, indicating efficient learning of the global model. Overall, the convergence analysis shows that algorithms designed to address client drift can significantly improve the stability and reliability of federated learning systems when training across distributed, heterogeneous datasets.

6.4 Communication Efficiency Comparison

The experimental results indicate that the communication efficiency varies across the evaluated algorithms. In the case of FedAvg, the model requires more communication rounds to achieve stable performance, particularly under Non-IID data conditions. This is primarily due to the absence of mechanisms that explicitly address the divergence of local updates. The FedProx algorithm improves communication efficiency by introducing a proximal term that stabilizes local training. By limiting the deviation between the local and global models, FedProx helps reduce the variance of client updates. This results in more consistent global model updates, allowing the model to reach stable performance with fewer fluctuations during training. The SCAFFOLD algorithm further enhances communication efficiency by using control variates to correct local gradient updates during training. This mechanism significantly reduces the variance caused by heterogeneous data distributions, enabling the global model to converge more reliably across communication rounds. As a result, SCAFFOLD typically achieves stable performance with fewer communication adjustments compared to the other algorithms.

6.5 Discussion of Observations

The comparative analysis of the three federated learning algorithms reveals several important insights regarding their performance in distributed medical image classification tasks. Under IID conditions across clients, all algorithms achieve strong performance. The results indicate that algorithms specifically designed to address the challenges of heterogeneous data distributions generally perform more reliably in federated environments.

The baseline FedAvg algorithm, while simple and widely adopted, may be unstable when client datasets differ substantially. This instability arises from divergent local updates, which can compromise the consistency of the global model aggregation process. In comparison, FedProx introduces a regularization mechanism that helps maintain closer alignment between local and global mod-

els during training. This additional constraint reduces the impact of local data heterogeneity and improves training stability. The results suggest that this approach can yield noticeable improvements in generalization performance when client data is unevenly distributed. Among the evaluated algorithms, SCAFFOLD demonstrates the most stable training behavior in heterogeneous settings. By incorporating control variates that adjust local updates, the algorithm effectively reduces the variance introduced by Non-IID data distributions. This leads to improved convergence stability and more consistent performance across communication rounds.

Overall, the experimental findings highlight the importance of addressing client drift and data heterogeneity in federated learning systems. The ability of an algorithm to manage these challenges directly influences its convergence behavior, communication efficiency, and overall predictive performance. These insights are particularly relevant for medical imaging applications, where data is naturally distributed across institutions and often exhibits significant variability.

Batch size also plays a crucial role as determines the

7 Practical Implications for Medical Federated Systems

The experimental analysis presented in the previous sections highlights how different federated learning algorithms behave under both IID and Non-IID data distributions. While these findings provide valuable insights from a methodological perspective, it is equally important to examine their practical implications for real-world healthcare environments. Medical federated learning systems must operate under strict privacy regulations, heterogeneous data distributions, and limited network resources. Therefore, selecting appropriate aggregation strategies and understanding system-level trade-offs becomes critical when deploying federated learning models in hospital networks.

7.1 Aggregation Strategy Selection for Healthcare Applications

One of the most important design decisions in a medical federated learning system is the choice of aggregation strategy for combining local model updates from participating clients. Different aggregation algorithms exhibit varying levels of robustness to data heterogeneity, convergence stability, and communication efficiency. The baseline FedAvg algorithm offers a simple, computationally efficient aggregation mechanism, making it suitable for scenarios where participating institutions maintain relatively balanced datasets. However, healthcare environments often involve significant variability in patient populations, disease prevalence, and imaging protocols. Under such heterogeneous conditions, the basic averaging strategy may lead to unstable convergence due to inconsistent local updates. To address these challenges, more advanced aggregation mechanisms such as FedProx and SCAFFOLD provide improved robustness in heterogeneous settings. FedProx introduces a proximal regularization term that constrains the

deviation of local models from the global model during training. This helps maintain alignment between client updates and reduces the impact of data imbalance. On the other hand, SCAFFOLD uses control variates to correct for the drift introduced by heterogeneous client updates, enabling the global model to converge more reliably. From a healthcare perspective, these mechanisms can be particularly beneficial because medical datasets collected across institutions rarely follow identical distributions. Selecting an aggregation strategy that accounts for such variability ensures that the resulting global model remains stable and reliable when trained on distributed clinical data.

Selecting an appropriate aggregation strategy depends on data heterogeneity, network infrastructure, and privacy requirements.

7.2 Trade-offs between Accuracy and Communication Overhead

While improving model accuracy is a primary objective of federated learning systems, practical deployments must also consider the communication overhead associated with distributed training. In federated learning, each communication round requires clients to transmit model updates to the central server and receive the updated global model. As the number of participating clients increases, this communication process can become a significant bottleneck. Algorithms that converge quickly with fewer communication rounds are generally better suited to large-scale deployments. Although methods such as FedProx and SCAFFOLD may introduce additional computational complexity during local training, they can potentially reduce the number of communication rounds required for convergence by improving training stability. This introduces a trade-off between computational complexity and communication efficiency. In practical healthcare systems, the optimal choice depends on the network infrastructure, available computational resources, and the scale of participating institutions. A well-designed federated learning framework must balance these factors to ensure efficient training while maintaining high predictive performance.

7.3 Deployment Considerations in Hospital Networks

One of the primary challenges in such environments is maintaining data privacy and security. There are many ways to implement privacy-preserving features, such as Differential Privacy, or to use blockchain-based platforms for secure sharing of model updates.[12]

Another important consideration is system interoperability. We need to create a platform where hospitals can collaborate on training sessions. This platform should be robust enough to dynamically manage the challenges faced during the Federated Learning Rounds. Ensuring seamless integration with these diverse systems is essential for practical deployment.

7.4 Summary of Findings

This study examined the application of federated learning techniques for the classification of chest X-ray images in a distributed medical environment. Three

federated optimization algorithms — FedAvg, FedProx, SCAFFOLD — were evaluated under both IID and Non-IID data distributions to analyze their effectiveness in handling IID & non-IID medical datasets. The experimental results demonstrated that federated learning can achieve strong classification performance while preserving data privacy across distributed healthcare institutions. All three algorithms were capable of learning meaningful patterns from the chest X-ray datasets. However, their behavior differed significantly when the data distribution across clients became heterogeneous. The analysis revealed that algorithms specifically designed to mitigate client drift, such as FedProx and SCAFFOLD, tend to provide more stable convergence in non-IID environments and improved generalization performance compared to the baseline FedAvg method in heterogeneous environments.

It was also observed that in IID environments FedAvg also performed arguably well.

7.5 Key Insights on Aggregation Behavior

The comparative evaluation of the three algorithms highlights the importance of selecting appropriate aggregation strategies in federated learning systems. The behavior of the aggregation mechanism plays a crucial role in determining the stability and reliability of the global model. FedAvg performs well when the data distribution across clients is relatively balanced, but it fails to achieve stability when there is significant heterogeneity across client datasets. FedProx addresses this limitation by constraining local model updates via proximal regularization, thereby improving stability in distributed training environments. SCAFFOLD further enhances convergence by introducing control variates that correct the direction of local updates. This approach effectively reduces the variance introduced by heterogeneous data distributions and enables more consistent model optimization across communication rounds. These observations emphasize that aggregation algorithms that mitigate client drift are particularly valuable for real-world federated learning deployments, where heterogeneous datasets are common.

7.6 Future Research Directions in Medical Federated Learning

1 This study demonstrates the potential of federated learning for medical image classification, several opportunities remain for further research and development. Future work may explore integrating privacy-enhancing technologies, such as differential privacy and secure multi-party computation, to further strengthen data protection in federated healthcare systems. Investigating personalized federated learning approaches could also help tailor models to the specific characteristics of individual institutions while still benefiting from global collaboration. Another promising direction is the development of adaptive aggregation strategies that dynamically adjust to varying data distributions across clients. Such approaches could improve the robustness of federated learning systems when deployed across large hospital networks. Finally, extending federated learning frameworks to incorporate multimodal medical data, including clinical records, imaging data, and

laboratory results, may enable the development of more comprehensive and accurate diagnostic models. By overcoming these challenges, federated learning is poised to serve as a core enabler of privacy-preserving, collaborative medical AI, paving the way for widely accessible healthcare diagnostic solutions.

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